Dexter Williams

PhD Student · Information

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Education

University of Wisconsin-Madison *Madison, WI*

PhD in Information *Aug. 2025 - Present*

- Advisor: Professor Jodi Schneider.
- Research Interests: Argument Mining, Computational Argumentation, AI and Law.
- 4.0 GPA

University of Illinois Urbana-Champaign *Champaign, IL*

PhD in Informatics *Aug. 2024 - July 2025*

- Advisor: Professor Jodi Schneider.
- 4.0 GPA

University of St Andrews *St Andrews, Scotland, UK*

BSc in Computer Science and Philosophy (Joint Honours) *Sep. 2018 - May 2022*

- First Class (3.9 GPA).

Research Experience

Project Assistant: University of Wisconsin-Madison *Madison, WI*

Supervisor: Professor Jodi Schneider *2025-Present*

- Support for a Second Edition of the Argumentation Mining monograph.
- Wrote an appendix for a book on argument mining, that documents existing corpora, and developed a website equivalent of the appendix.
- Acquired copyright permissions for the 100+ figures used in the book.
- Developed a data pipeline that measures the agreement of databases one the retraction status of scientific articles.

Research Intern: Boise State University *Boise, ID*

Supervisor: Professor Casey Kennington *2025*

- Joined the Retico team at Boise State University for 1 month to aid in the research and development of state-of-the-art incremental processing systems.
- Developed two new modules for Retico to handle speech diarization and automatic speech recognition.

Research Assistant: University of Illinois Urbana-Champaign *Champaign, IL*

Supervisor: Professor Jodi Schneider *2025*

- Provided feedback for a new edition of a book on argument mining.
- Read and categorized 100+ papers on argument mining to assist in the writing of a new chapter.
- Funded by the University of Illinois at Urbana-Champaign Campus Research Board, $14,714.

Research Assistant: University of Illinois Urbana-Champaign *Champaign, IL*

Supervisor: Professor Bertram Ludäscher *2025*

- Joined the explainable AI through computational argumentation (XAI-CA) project, which explores how argumentation can aid in explainability.
- Applied relation-based argument mining to US-legal documents to automatically construct structured argumentation frameworks.
- Funded by the 2025 Cardiff University-University of Illinois System Joint Research and Innovation Seed Grants Program.

Undergraduate Research Assistant: University of St Andrews *St Andrews, Scotland, UK*

Supervisor: Dr. Kasim Terzić *2019-2020*

- Developed a program used by the Marine Biology department to automatically identify seal populations in aerial photographs.

Professional Experience

Software Engineer/Consultant *Bath, England, UK*

Capgemini Engineering *2022-2024*

- Joined the development team for a large rail project.
- Writing and testing the formal specification of the project.
- Implemented specification in Ada SPARK.
- Tested the project using a suite of tools developed under the HICLASS Research Programme.

Data Science and Innovation Intern *Leeds, England, UK*

National Health Service (NHS) Digital *2021*

- Joined the reproducible analytical pipelines team at NHS Digital for ten weeks.
- Produced four guides that teach good software engineering practices.
- Investigated existing PySpark libraries to find a simple and reproducible way for teams to implement basic metadata input validation in their projects.

Teaching Experience

Teaching Assistant: University of Illinois Urbana-Champaign *Champaign, IL*

Information Science 203: Analytical Foundations for Information Problems *2024-2025*

- Two semesters teaching 80+ students introductory formal logic, discrete mathematics, and statistics.

Publications

Conference Papers

Williams, D. (2026). That feels convincing: Assessing the quality of pathotic arguments. Argumentation in the Digital Society. Proceedings of the 5th European Conference on Argumentation (ECA 2025). Forthcoming.

Williams, D. (2026). Decorum in the forum: Emotional interactions in U.S. presidential press conferences. Arguing Democracy. Proceedings of the 2025 NCA/AFA Argumentation Conference (Alta). Forthcoming.

Zheng, H., **Williams, D.**, & Ludäscher, B. (2025). Modeling U.S. Supreme Court briefs with computational argumentation. In R. Markovich, L. D. Caro, A. Rapp, & C. Schifanella (Eds.), Legal Knowledge and Information Systems. SAGE Publications. https://doi.org/10.3233/FAIA251631

Workshop Papers

Stade, C., **Williams, D.** (2026). Retraction without traction: Towards post-publication invalidation in open science infrastructure. Proceedings of the 6th International Workshop on Scientific Knowledge Representation (Sci-K).

Williams, D., Liu, S., Stede, M., Wachsmuth, H., & Schneider, J. (2026). AMResources: Cataloging argument mining datasets. In M. Elaraby, A. Hautli-Janisz, J. Romberg, E. Musi, F. Ruggeri, & J. Lawrence (Eds.), Proceedings of the 13th Workshop on Argument Mining and Reasoning (pp. 52–58). Association for Computational Linguistics. https://doi.org/10.18653/v1/2026.argmining-1.5

Zheng, H., **Williams, D.**, & Ludascher, B. (2025). Using LLMs to model arguments in U.S. Supreme Court briefs: Preliminary report. Proceedings of the International Workshop on Translating Natural Legal Language into Formal Representation (NLL2FR 2025).

Non-Refereed/Lightly Refereed Publications, Presentations & Posters

Williams, D., & Schneider, J. (2026). Emotional argumentation in Second Look. Talk at The 2026 Symposium of the Applied Rhetoric Collaborative.

Williams, D., & Hatt, A. (2026). Don't argue with cops? The harms of "fighting words". Poster at the University of Wisconsin-Madison May Day Tea 2026.

Fellowships, Awards, & Honors

2026 **Humane Studies Fellowship**, Institute for Humane Studies (IHS021082) *$ 3000*
2026 **Best paper award**, International Workshop on Scientific Knowledge Representation *€ 300*
2024 **Informatics Recruitment Fellowship**, University of Illinois Urbana-Champaign *$ 30,000*
2025 **iSchool Conference Presentation Award**, University of Wisconsin-Madison *$ 1,612*
2025 **Informatics Conference Presentation Award**, University of Illinois Urbana-Champaign *$ 1,000*

Mentoring

2025-2026 **Undergraduate Independent Study**, Supervisor, University of Wisconsin-Madison
2021 **St Andrews Mentoring Initiative**, Mentor, University of St Andrews
2020 **Coronavirus Tutoring Initiative**, Tutor

Service

Peer Reviewer

11th International Conference on Computational Models of Argument (COMMA 2026)
15th edition of the Language Resources and Evaluation Conference (LREC 2026)
13th Workshop on Argument Mining (ArgMining 2026)
12th Workshop on Argument Mining (ArgMining 2025)